

Quadratic Forms and Their Distributions

Every sum of squares in a linear model is a quadratic form $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ in a normal vector, and every F and t statistic is a ratio built from such forms. This chapter works out their laws: the noncentral chi-squared, F and t distributions and the monotonicity of their tails; the mean, variance and a canonical representation of an arbitrary quadratic form; exactly when a quadratic form is chi-squared; when two forms, or a form and a linear function, are independent; and Cochran's theorem, which says when a sum of squares splits into independent chi-squared pieces.

What you need. [Chapter 1](#) (spectral theorem, symmetric idempotent matrices, trace), [Chapter 2](#) (means and covariances of random vectors) and [Chapter 3](#) (the multivariate normal distribution and its independence properties). Orthogonal projections appear here only as symmetric idempotent matrices. [Chapter 6](#) develops their geometry.

4.-1.1 Roadmap

- [Noncentral chi-squared, t and F distributions](#)
- [Mean and variance of quadratic forms](#)
- [When a quadratic form is chi-squared](#)
- [Independence of quadratic forms and linear forms](#)
- [The Cochran theorem](#)
- [Summary and notes](#)